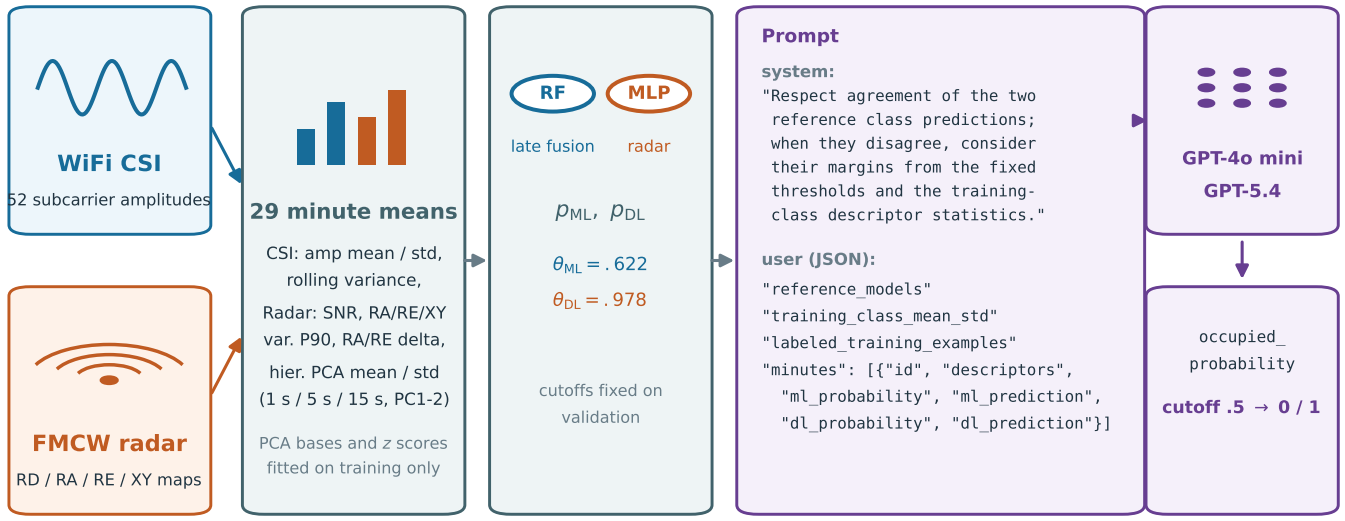
WiFi CSI
52 subcarrier amplitudes
FMCW radar
RD / RA / RE / XY maps
29 minute means
CSI: amp mean / std, rolling variance,
Radar: SNR, RA/RE/XY var. P90, RA/RE delta,
hier. PCA mean / std (1 s / 5 s / 15 s, PC1-2)
PCA bases and z scores fitted on training only
RF
MLP
late fusion
radar
p_{ML}, p_{DL}
$\theta_{ML} = .622$
$\theta_{DL} = .978$
cutoffs fixed on validation
Prompt
system:
"Respect agreement of the two reference class predictions; when they disagree, consider their margins from the fixed thresholds and the training-class descriptor statistics."
user (JSON):
"reference_models"
"training_class_mean_std"
"labeled_training_examples"
"minutes": [{"id", "descriptors",
"ml_probability", "ml_prediction",
"dl_probability", "dl_prediction"}]
GPT-4o mini
GPT-5.4
occupied_probability
cutoff .5 → 0 / 1